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Differential privacy for renewable energy resources based smart metering

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HIGHLIGHTS

- We discuss integration of differential privacy in renewable energy based smart grid.
- We provide importance of privacy in renewable energy resources based smart meters.
- We provide analysis of privacy preservation strategies implemented in smart grid.
- We presented simulation work over real-time RER based smart metering data.
- We evaluated that our DPLM algorithm outperforms previously proposed strategies

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ABSTRACT

The increasing energy costs and increase in losses in traditional power grid system triggered the integration of Renewable Energy Resources (RERs) in smart homes. The global desire of consumers to rely on RERs such as solar energy, and wind energy has increased dramatically. Similarly, the IT technologies are also playing their part in smart grid development, such as real time data monitoring. On the other hand, with the advancement of these IT technologies in smart meters, the privacy of customers is also at risk Smart grid utility knows the exact generation of any specific renewable resource in a specific interval of time. Utility need to monitor this real time data for load forecasting and implementation of demand response scenarios. However, the utility may misuse the data and may increase the prices for specific time slots when RERs are not present. Similarly, real time monitoring of data can lead to estimation of life routines of users such as sleeping habits, time of usage of heavy appliances, and lifestyle. In this paper, a Differential Privacy based real time Load Monitoring approach (DPLM) is proposed that preserve the privacy of users by masking the values of load in such a way that utility will not be able to judge the usage of specific RER and the daily routine of any smart meter user. We compare our scheme with Gaussian Noise Differential Privacy (GNDP) strategy. Experimental results validate that our DPLM approach provides a desirable solution to protect smart grid user's privacy by efficient noise addition and peak value protection along with having an error rate of only 1.5%.

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1. Introduction

The increase in the cost of fossil fuel-based energy generation has developed an interest in the integration of different forms of Renewable Energy Resources (RERs) for domestic use [23]. RERs are energy resources that can be used to generate additional energy in order to serve the needs of a specific area or building. RERs can be of diverse nature, but the traditional RERs which are used over homes include solar panels, small wind turbines, and small biomass plants. This integration of RERs in smart grid is one of the future direction of energy systems because of their heterogeneous nature and dynamic capabilities [39]. In future, it is predicted that smart homes will be equipped with more than one RER including solar, and wind energy at the same time [11]. However, due to the intermittent nature of RERs (e.g. available only in specific hours), they are generally used in combination with traditional fossil fuel based energy from grid utility [39].

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Along with this enhancement, advanced information and communication technologies (ICT) have become an essential part of smart grid architecture [5]. Two-way communication of energy and data in smart grid is paving its path towards more intelligent, secure, and automated control of smart grid [12]. Smart meters and Advanced Metering Infrastructure (AMI) networks are transmitting real time data of smart homes to grid utility after a specific interval of time for future demand response management and load forecasting strategies [20,24]. This data can help utility to make an estimate regarding futuristic use of energy in order to overcome any load shedding or power cut. Similarly, this fine-grained data can also lead to an early detection of any sort of theft or tempering in smart meter.

However, this reported smart meter data can be used for several financial and commercial purposes. For instance, smart grid utility can increase prices of certain hours in which RERs are not available, or this data can be sold out to business corporates who can use this data to carry out targeted advertisement for their potential customers. Furthermore, the data collected by smart grid utility can also cause serious harm to privacy of grid users [5,21]. Real time monitoring of energy data can reveal private information regarding daily routine, habits, and lifestyle of customers. This can simply be done from data by using basic Non-Intrusive Load Monitoring (NILM) techniques [1]. In the same way, real time monitoring can disclose private information about presence and absence of any particular RER in specific hours, hence utility can exploit this information by certain ways (e.g. may increase the prices of electricity in certain hours of the day). Therefore, preserving privacy of smart meter data along with real time monitoring is the need of future energy systems. Several privacy preserving techniques of smart meters are presented in previous literature. These techniques include battery-based load hiding (BLH) [50], cryptographic encryption [14], data perturbation [18], splitting of data into different chunks before transmission [10], trusted third party [27], and decentralized framework for data aggregation [5]. A comprehensive privacy protection strategy named as differential privacy was proposed by C. Dwork in 2006 [15]. Differential privacy works over the principle of perturbing data by adding adequate amount of noise [22,45]. However, this scheme also proved to be fruitful in various real-time applications as well. Detailed information about differential privacy is presented in Section 3.1.

In this paper, we present a real time differential privacy load monitoring (DPLM) algorithm for smart meter users integrating RERs. This technique provides a desirable level of privacy to smart meter users by first accumulating all values of energies, adding noise, and finally restricting values to a specific peak limit before transmitting it to utility. Along with this, we integrate the functionality for monthly monitoring, this keeps the track of total usage of grid energy, solar energy, and wind energy till end of the month. These accumulated values provide statistical information about the total grid energy usage in the month (for billing purpose) and total solar and wind energy generated (for future demands response and load forecasting scenarios).

1.1. Comparison with previous literature

Many previous researches have been carried out to use differential privacy in order to protect privacy of smart meters. Few researches use the concept of battery load balancing and its integration with differential privacy to provide dual perturbation, such as the authors in [48,50] used battery load hiding along with differential privacy to protect mutual information sharing. Similarly, another approach is to use differential privacy directly to perturb smart metering data, researchers such as in [3,7,17,49] used various mechanisms of differential privacy noise addition to protect smart meter user privacy according to their requirement. A detailed description about these previously proposed strategies and their comparison with DPLM is provided in Section 2. To conclude, differential privacy is a well-researched topic in the field of smart grid, however the researches have not yet considered the integration of differential privacy in renewable energy resources based load monitoring domain. Our proposed DPLM mechanism will be the pioneering step towards integration of differential privacy based renewable energy resource based smart meter reporting.

1.2. Contribution of this article

The novelty and motivation of our proposed strategy is first described here:

- Privacy preservation for smart meters in literature have been carried out mainly by considering battery load balancing, data perturbation, and encryption strategies [14,50]. These schemes have considered and optimized various applications of smart grid to benefit smart meter users as mentioned in Table 1. However, privacy protection strategy have never been employed for RERs based smart grid scenario. To the best of our knowledge, our novel contribution is the development of differential privacy strategy in renewable energy resources based smart grid.
- To optimize privacy a bit further, we introduced a peak factor in our differential privacy based privacy preservation strategy. Selecting the optimal peak-threshold value according to environmental scenario is difficult in smart grid. This selection of optimal peak-threshold becomes even more difficult in availability of multiple RERs. Owning to these limitations, we have developed differential privacy preservation strategy with suitable peak-threshold value on which our strategy protects RERs real-time data efficiently.
- In literature, various analytical differential privacy models for other smart grid scenarios have been formulated, but analytical modeling for RERs based smart metering considering peak-threshold preservation has merely been touched. We formulated analytical model by developing the algorithm to protect reporting of real-time smart meter and RERs values.

The main contributions of this paper are as follows:

- We provide a detailed overview and analysis of privacy preserving strategies of smart meters.
- We preserve privacy of daily routine, habits, and lifestyle of RERs integrated smart meter users.
- We preserve information about intermittent (only available for specific hours) availability or unavailability of renewable energy resources.
- We develop an algorithm for monthly accumulation of grid and renewable energy that only transmits the values for monthly billing and transmit preserved energy value after every 10 min.

To the best of our knowledge, no literature work that protects the intermittent readings of RERs along with protecting peak-threshold by using differential privacy have been discussed in the past.

1.3. Organization of the paper

The remaining sections of the paper are organized according to following pattern, in Section 2, we give an overview of related work that is carried out so far. The contents of Section 3

Table 1
Comparative view of privacy preservation techniques in smart grid energy systems with their specific technique, enhanced parameters, privacy criterion (ε, δ), computational complexity, and experimental platform.

Main category	Ref no.	Name of privacy mechanism	Privacy technique used	Enhancement due to proposed mechanism	Platform used	Privacy criterion	Time complexity
Encryption	[35]	Private data collection for AMI networks	Hashing & cryptographic encryption is used for data collection	• Reduced computational overhead • Optimized packet size	NS-3	N/A	–
	[6]	AMI data aggregation while preserving privacy	Homomorphic encryption used for smart meters	• Network delay enhancement • Reduced network traffic	OPNET 14.5	N/A	$O(n * M(n))$
	[2]	Light-weight smart grid data aggregation	HAN data privacy by Lattice-based homomorphic encryption	• Optimized computational overhead	MATLAB	N/A	–
Battery Load Hiding	[48]	DP battery supported smart metering	DP in-conjunction with MAB algorithm and battery load	• Reduction in extra cost • Enhanced mutual information sharing	N/A	(ε, δ)-differential privacy	–
	[50]	Battery load Hiding-Exp3	Context aware DP based battery load balancing	• Enhanced event detection accuracy Mutual information sharing	N/A	(ε, δ)-differential privacy	$O(log\ n)$
Direct Data Perturbation	[3]	DREAM	Gamma distribution for privacy preservation & encryption to secure data	• Cluster error reduction • Appliance multi-slot privacy	Electricity Trace Simulator	ε-differential privacy	–
	[17]	DP for real smart metering data	Concept of point-wise DP with Laplacian noise is used	• Signal smoothing • Timestamp privacy	N/A	ε-differential privacy	–
	[49]	DP cost-friendly (CDP) strategy	Cost consideration with DP and MAB algorithm is used	• Static & dynamic reporting • Mutual information sharing	N/A	(ε, δ)-differential privacy	$O(log\ n)$
	[7]	DP with appliance usage in smart metering	Laplacian noise is used to mask critical data	• Real-time data privacy • Demand response calculation	Simulators[2] for C	ε-differential privacy	$O(1)$
	This Work	DP for renewable energy resources based smart metering	Laplacian noise along with peak-factor limitation is used for RERs scenario	• Instantaneous data reporting • Peak value preservation • Renewable resources instantaneous reading	Python	ε-differential privacy	$O(n)$

contain the description of our proposed DPLM strategy in RERs based smart grid. In Section 4, performance evaluation of DPLM strategy is presented with help of experimental results. Finally, in Section 5, conclusion and future research directions are described.

2. Related work

2.1. Background

The term smart grid is a modernized version of traditional energy grid of 20th century [19,29]. Traditional system of energy grid was only used to carry out basic electricity generation and transmission tasks. However, the modern smart grid can be used to perform countless functionalities because of its bidirectional information and electricity flow feature [30]. Smart grid architectures are capable of providing efficient communication, and data storage by using modern communication technologies such as ZigBee, power line communication, wireless LAN, and cognitive radio communication [25]. Taking a step further, researchers have also started the integration of renewable energy in smart grid architecture, specifically in smart homes for efficient green renewable energy. This integration of RERs with smart homes is credited as one of the most beneficial enhancement of smart grid as it can be used to cope up with power shortfalls along with efficient improvement in reliability and economy [9,36]. Smart grid is proved to be beneficial by improving our energy systems, but it also comes up with various issues and the most critical among them are security and privacy threats of smart meters.

In a smart grid environment, smart meters are connected with each other and with the grid utility via various communication mediums, this connection constitutes to form a complete metering network called as advanced metering infrastructure. In this AMI scenario, smart meters report their real-time fine grained data from smart homes to grid utility in order to perform certain tasks such as load forecasting, demand response calculation, etc. [4,13,47]. However, this accurate data can reveal the personal life of individuals of that smart homes, as many NILM techniques have been developed in order to reveal exact usage of appliance from smart meters data [46]. This data can be used to construct exact usage patterns of smart homes, which will reveal the presence or absence of smart home users in a specific interval of time, which afterwards can be used by any adversary as well. In order to protect this privacy leakage, numerous techniques have been proposed by researchers in smart grid scenario. Two major techniques used for such cases are encryption and data perturbation.

2.2. Privacy preserving techniques in smart meters

Encryption is a strategy in which public and private keys are used by smart meter users to protect personal data from adversaries. In encryption, a set of public key is shared with every smart meter node, while only the grid utility has access to private keys of each smart meters via which it decrypts the reported reading. In [35], Mohammed et al. proposed an efficient privacy protection scheme for AMI networks by using the concept of symmetric-key-cryptography. Furthermore, they used

the concept of hashing in order to mask smart meter values during data collection from an AMI network. In order to reduce computational overhead, the authors used key renewal procedure after a long interval of time. Furthermore, packet size is optimized and verified by authors by performing simulations over NS-3. Moving a step further, authors in [6] proposed a data aggregation strategy by focusing over the concept of homomorphic encryption by using OPNET 14.5 simulator. The proposed strategy reduces latency, delay, and network traffic in smart metering network along with preserving the private smart meter data from intruder. Another homomorphic encryption based privacy preservation scheme by considering data aggregation is proposed by Abdallah and Shen in [2]. The authors used the concept of lattice-based homomorphic encryption to reduce computational overhead of the network and to preserve the consumer privacy. In the proposed strategy, the appliances used in smart home self-aggregate their consumption without the involvement of smart meter, in order to prevent the leakage of appliance usage pattern from adversaries. Encryption techniques provide a suitable privacy protection from third party intruders, but in context of practical implementation, these algorithms are computationally complex [33,38], and sometimes they require the interconnection of every smart meter in the area for generation of distributed and secret keys. Therefore, if one meter gets failed in the network of n number of meters, then the aggregation of data becomes impossible due to missing distributed keys in the network [7].

Battery load balancing is another approach proposed by researchers to protect smart meter user privacy. In this type of approach, an external battery is used to balance the load in order to prevent smart grid utility from knowing the actual behavior of smart home users. The authors in [48] used the concept of battery controller to switch between grid and battery and then worked over addition of noise to preserve extra privacy. The authors focused over reduction of extra cost along with enhancement in mutual information sharing. Similarly, Zhao et al. in [50] proposed a strategy to prevent data disclosure by using the merger of differential privacy with battery load balancing. The authors enhanced event detection accuracy and presented a context aware battery load balancing, that proved to be an optimal battery-based solution for smart meter users. However, along with the less lifespan of a battery, the cost and environmental effect of batteries cannot be ignored [7]. The authors in [34] compared the cost of batteries according to their approximate lifespan of two years and demonstrated that battery load balancing is not a low cost solution. Thus, using battery to protect privacy is not a suitable solution because of its cost and low life.

Privacy protection using direct data perturbation concept of differential privacy is another strategy that is used by many researchers in order to protect smart meters privacy. In this approach, direct data from smart meter is perturbed by adding appropriate amount of noise. The first step towards this strategy was taken by Acs et al. in [3]. In the proposed strategy, the authors used gamma distribution to calculate the noise and then perturbed smart meter data before transmitting it to utility. The authors further protected appliance multi-slot privacy along with reducing cluster error. Another work in this field is carried out by Eibl and Enger in [17], the authors introduced the concept of point-wise differential privacy by perturbing real-time smart metering data using Laplacian noise. Furthermore, the authors smoothened the dispersed signal and also preserved timestamp privacy by hiding actual data. In [49], a cost-friendly differential privacy strategy is proposed for smart metering network by using MAB algorithm. The major focus of article was to preserve static and dynamic reporting of smart meter data along with enhancing mutual information sharing and reducing the cost. Furthermore, the authors in [6] used data perturbation to preserve appliance usage, and perturbed appliance readings so that adversary may not be able to target or infer the exact readings. Till now, differential privacy comes out as one of the most promising solution to protect smart metering privacy.

The above mentioned articles applied differential privacy and other privacy preservation strategies in various smart grid applications. However, in literature, differential privacy has not been integrated in smart meter that work in conjunction with RERs. This paper proposes the technique to preserve instantaneous energy generation from renewable energy resources, along with the protection of daily life routine and peak value reporting of smart meter users by using differential privacy concept.

3. Differential privacy based load monitoring (DPLM) strategy in RERs based SG architecture

From smart meter real-time data, one can easily infer the lifestyle and RERs usage of any consumer [8]. For example, inhabitants' daily routine, time of usage of heavy appliance, and electricity generation timings of RERs can be easily deducted from basic smart meter data. Similarly, data techniques in smart meter can make estimate that any particular appliance is not working at a desired level of efficiency and needs to be replaced. This specific usage can be obtained from the collected data of smart meters by using NILM techniques [1].

Furthermore, this data can be used for various commercial and financial benefits. For example, utility can raise the prices of electricity for certain hours when energy from RERs is not available. Similarly, utility can sell this data to business corporates that can do targeted advertisements and can target their potential customers. In order to do so, various marketing companies gather data collected from smart meter and try to extract their electricity usage along with exact time. This data is further refined using NILM techniques and from the load values, it can be estimated that whether some appliance (e.g. toaster, washing machine, etc.) is working on optimal condition or not. Similarly, it can also be estimated that a specific appliance is going to run-down very soon, in this way these companies can target the specific house for the appliance of they might will require in near future. Another way that these companies do adopt is to target specific hours in which all residents are available in house and doing something particularly related to TV or electronic media. In this manner, they can target maximum audience and can increase the impact of their advertisement. Likewise, any intruder can plan any illegal activity in the house after estimating the routine of consumers. In context of the usefulness of real-time data, the utility needs real time monitored data to identify energy quality, monthly billing, load forecasting optimization, and developing of demands response plans. Similarly, this data can be used to improve management of network, calibration of grid stations, reduction of on-peak loads, and can also be used to facilitate the consumers in various ways. Therefore, collection of real-time data is critical for smart grid utility. But on the other hand, privacy of customers cannot be compromised. This is one of the major reason for denial of implementation of smart meters by few residential customers [26].

Considering in view all above discussion, we develop a protection mechanism that preserves the desired level of privacy of smart meter users and also fulfill the requirement of real-time data monitoring by utility. In this section, differential privacy mechanism, design goals, system model, adversary model, and noise addition mechanism of our DPLM approach is discussed.

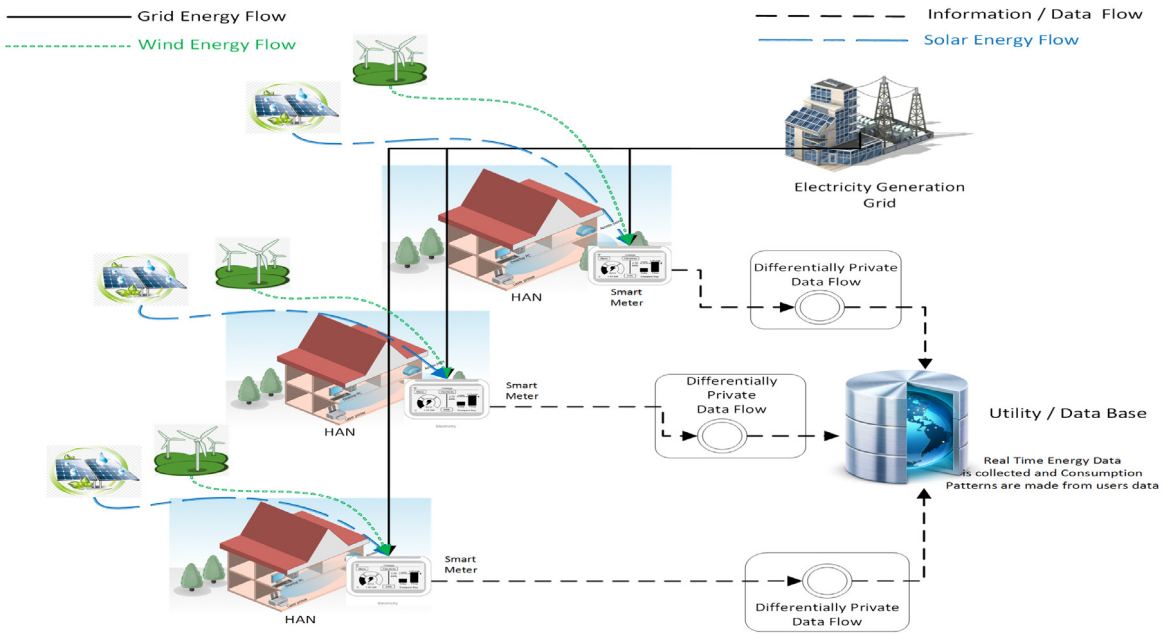


Fig. 1. Application scenario of smart homes and grid utility describing functioning of differential privacy in Renewable Energy Resources (RERs) based smart grid. In this scenario, smart meter is integrated with every smart home that communicates with utility and transmits real-time readings after every 10 min.

3.1. Differential privacy

Differential privacy was first introduced by C. Dwork for statistical datasets [15]. It stated that a mechanism is said to be differentially private if its output does not affect significantly by addition or removal of a single participant of dataset.

Definition 1 (*Adjacent Datasets*)**.** In an n-dimensional data set M^n, a randomized query function R provides ε-differential privacy P_{r1} if $\forall c1, c2 \in M^n$ differ only in one element and all values of $S \in range(R)$. Where M is the dataset, S is the output, and R is the function that satisfies ε-differential privacy [43].

$$P_{r1}[R(c1) \in S] \leq e^{\varepsilon} \times P_{r1}[R(c2) \in S] \tag{1}$$

Where $range(R)$ denotes the range of output of function R.

The value of ε is the privacy parameter value and to achieve good privacy, a smaller value of ε is desired [32].

Definition 2 (*Sensitivity*)**.** Suppose that $f(.)$ is a given query function in the provided dataset. Then the value of global sensitivity Δf can be denoted as δf_s or δf [40]. The formula is given in the equation below:

$$\delta F_s = \max_{c1,c2} |f_s(c1) - f_s(c2)| \tag{2}$$

$\forall c1, c2$ differs in only at most one element.

In our scenario presented in Fig. 1, the output of real-time smart meters data differs from traditional data sets because we are considering and perturbing real-time values after every instance of time. Similarly, the value of query function $f(.)$ can just be the sum of smart meter value (X_c) at a particular instant of time "t" [17]. This can be demonstrated as follows:

$$\delta f = \max_{c} \sum_{t=1}^{T} |X_{c,t}| \tag{3}$$

The above equation describes the global sensitivity value when we consider a single point-wise input reading.

Definition 1 (*Point-Wise Sensitivity*)**.** Eibl et al. [17] evaluated the concept of point-wise privacy in real-time smart metering data and proposed a theorem to preserve privacy. He stated that each time print can be considered independently from other points and can be perturbed individually by utilizing the application of Laplacian Mechanism. The above mechanism is called as the single later version of differential privacy, in contrary to the vector version of Eq. (2). The new global sensitivity can now be defined as follows:

$$\delta_{point\,wise}(f) = \max_{t,c1,c2} |f_t(c1) - f_t(c2)| = \max_{c,t} |X_{c,t}| \tag{4}$$

Differential Privacy in case of smart metering data follows the basic theorem of Eq. (4). Therefore, the privacy leakage of value of ϵ at a given time point t in the equation is chosen to be $\varepsilon_t = \varepsilon/t$.The desirable leakage value from definition is usually between 0 and 1 ($\varepsilon \leq 1$), but for point-wise noise calculation we set $\varepsilon = 1$. The detailed explanation about calculation and addition of noise in our approach is demonstrated in Section 3.6.

3.2. Design goals

The previous approaches of differential privacy in the literature only considered grid energy perturbation, but none of them considered preserving of information of generation of RERs. The required design goals of DPLM strategy are given as follows:

- Keeping record of total RERs energy generation from solar and wind sources, along with aggregation of total energy consumed from grid.
- Achieving data disclosure privacy by using Laplacian noise addition characteristic of differential privacy with maximum 10% error rate.
- Preserving privacy of real-time energy usage by reporting the perturbed values of smart meter after addition of Laplacian noise.
- Preserving RERs privacy by preventing the reporting of instantaneous value of generation from a specific RER.
- Protecting peak-instantaneous load reporting for a specific interval by normalizing the value and adding the remainder to next reading.

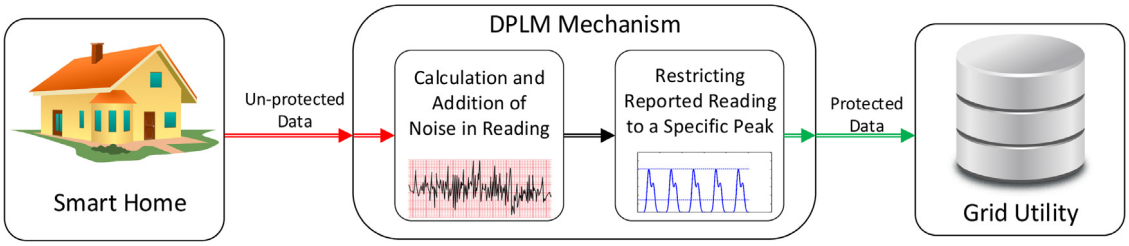


Fig. 2. An Illustrative Demonstrating of Functioning of DPLM Privacy Protection Strategy.

Algorithm 1 Algorithm for Differential Privacy based Load Monitoring in Renewable Energy Resources based Smart Grid.

while DPLMinRERsbasedSG **do**
  Input ← N(No.OfReadings), F(Function), SE (SolarEnergy), WE (WindEnergy), GE (GridEnergy), MeterID, TimeSlot
  Output ← TGE(TotalGridEnergy), PE (Percentage Error), $TN_{oisy}E$ (TotalNoisyEnergy), TSE(TotalSolarEnergy), TWE (TotalWindEnergy), TRE (TotalRenewableEnergy), SN (SummedNoiseEnergy)

  FUNCTION → $DPinRERs(N, F, SE, WE, GE, SE)$
  while Data received from RERs **do**
    $P \leftarrow$ Maximum Peak Load that can be transmitted
    Aggregate total grid energy $(TGE = TGE + GE)$
    Aggregate total solar energy $(TSE = TSE + SE)$
    Aggregate total wind energy $(TWE = TWE + WE)$
    Sum up three energy values $(Sum = SE + WE + GE)$
    Aggregate total energy $(TE = TE + Sum)$
    if $sum > 0$ **then**
      Call DiffNoiseCalulation (N, Xi, f)
      Add Differential Noise to Sum $(SN = Sum + Noise + EX)$
      if $SN > P$ **then**
        $EX = SN - P$
        $SN = P$
      else
        $EX = 0$
      end if
    else
      $SN = Sum$
    end if
    Aggregate Total Sum with Noise $(TN_{oisy}E = TN_{oisy}E + SN)$
    Calculate Energy from Renewable Resources $(TRE = TE - TGE)$
    Calculate Percentage Error $(PE = (TN_{oisy}E - TE))$
    Return $SN, TGE, TSE, TWE, TN_{oisy}E, PE, TRE,$
  end while

  FUNCTION → $DiffNoiseCalulation(N, Xi, f)$
  Calculate sum of monthly data for each house
  while noise needs to be added **do**
    Find maximum allowed error $(MaximumError(ME)(10\%))$
    Calculate S.D and Variance $(\sigma_{\varepsilon 0}), (\sigma^2_{\varepsilon 0})$
    Calculate Scale of Laplacian Noise (b)
    Find value of maximum noise $(MaximumNoise(MN))$

    FUNCTION → $LaplacianNoise()$
    Calculate Laplacian Noise $f(x; \mu, b)$
    Return $noise_t$
  end while
end while

3.3. Privacy requirements for RER based smart metering

Designing differential privacy based approach for RER based smart meter possess certain design criteria that needs to be considered. Traditional privacy preserving approaches designed for basic smart meters do only work protect time of use (ToU) data in order to preserve appliance usage and real-time pricing. However, integrating RER with smart meter adds up plenty of other constraints that require considerable attention such as solar panel real-time generation, wind turbine real-time generation, demand side management data, load scheduling information, etc. [39]. For example, grid utility can do dynamic pricing and can increase prices at the time of absence of RERs, similarly, the amount of energy generated from RERs can also make social competition among smart meter users. Therefore, specific privacy preserving strategy for RER based smart meter is required to be developed in order to cope up with the increasing demand of privacy of smart meter users.

3.4. System model of DPLM

In this section, the representation of integration of RERs with smart meter and grid utility is discussed. The scenario diagram of our proposed architecture is presented in Fig. 1. In the given figure, each smart home is connected with solar panels and small wind turbines for generation of renewable energy, it is also connected with electricity grid power for balancing of energy during absence of RERs energy. Similarly, each home has a smart meter that ensures its real-time communication with the grid utility. Smart meter gathers information from different energy resources, aggregate the required energies, calculate Laplacian noise add the noise if it is required, restrict the values to a specific peak and then transmit it to utility by using any wireless medium. The complete flowchart for this process is shown in Fig. 3, which shows the accumulation of all energies along with steps of our proposed strategy. First critical step after energy accumulation in Fig. 3 is to check whether the sum of energies is greater than 0 or not, and if it is not greater than 0, then the steps for noise addition were not called in order to save the computational power in case of no one using any energy in smart homes. Afterwards, the flowchart demonstrates the steps of calculating error value, variance, and noise value which are carried out according to phenomenon explained in Algorithm 1. Finally, the accumulated value is compared with the maximum peak and is adjusted accordingly to transmit to utility. Additionally, a brief illustrative example of differential privacy based load monitoring for renewable energy resources based smart metering is provided in Fig. 2. In the figure, the mechanism is divided into three stages, at the first stage, the unprotected data from smart meter is taken and sent to DPLM mechanism. In the second stage, DPLM adds noise and restricts the transmitted value to a specific peak. Finally these protected values are transmitted and then stored in smart grid utility at stage 3.

Furthermore, at end of a specific time period (e.g. monthly period), the smart meter calculates the total grid electricity usage (for billing), total monthly solar and wind energy generated (for demand response and load forecasting), and percentage error for calculation of accuracy of real-time readings. The complete algorithm explaining the functionality of DPLM strategy in RERs based SG is given in Algorithm 1. The time complexity for Algorithm 1

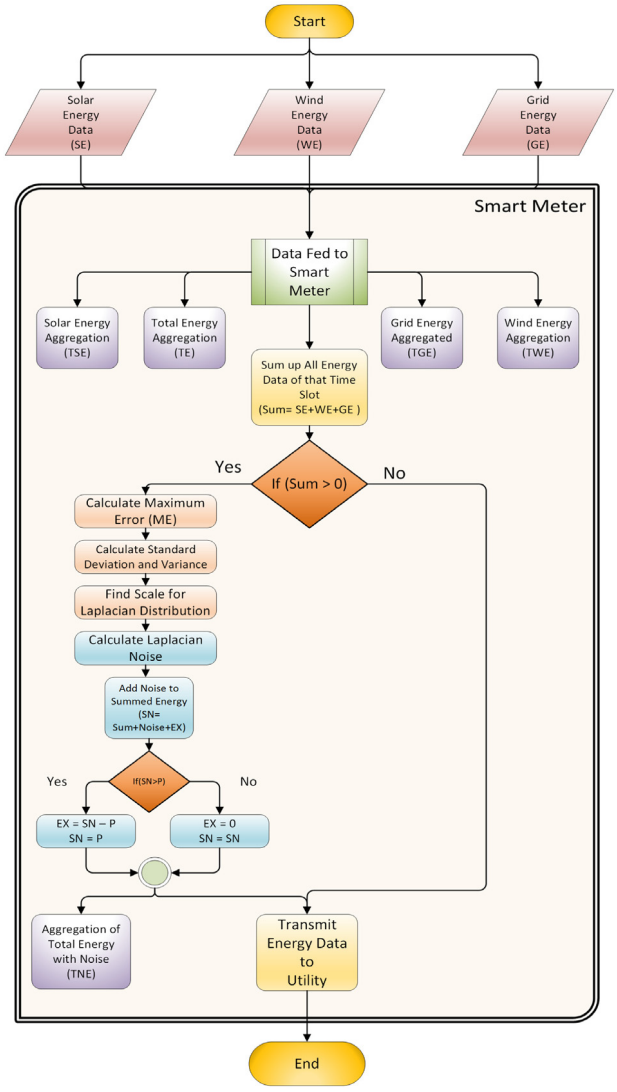


Fig. 3. Flow chart showing implementation of differential privacy in renewable energy resources based smart grid.

is $O(n)$ because it iterates only once for all smart meter readings. Some key notations used in our algorithm is presented in Table 2.

3.5. Adversary model

The adversary can be any intruder trying to infer the real-time readings "$X''_{c,t}$ of smart meter. Adversaries are actually interested to know the lifestyle and instantaneous renewable energy generation from a specific smart home. In this paper, we consider two types of adversaries/attacks [42]: (1) external attack, in which eavesdropper tries to compromise the communication link between smart meter and grid utility and tries to take advantage of the information being transmitted in between; (2) internal attack, in this type of attack grid utility/aggregator acts as adversary and tries to misuse the collected information of smart meters, this adversary can use this smart meter data itself or can further provide it to advertising companies that further can misuse this data to do targeted advertisements. Adversary is assumed to be curious-but-honest, as it can only obtain the smart meter data for usage patterns but will not alter, delete, insert, or modify the smart meter readings.

Table 2
Key notations and their terminology.

Notation	Meaning
b	Scale of Laplacian Noise
EX	Excessive energy for next iteration
ε	Privacy controlling parameter
σ_{ε_0}	Standard deviation
$\sigma^2_{\varepsilon_0}$	Variance
SE	Solar energy
WE	Wind energy
GE	Grid energy
MN	Maximum noise value
TSE	Total solar energy
TWE	Total wind energy
TGE	Total grid energy
TRE	Total renewable energy
N	No of readings
SN	Summed noise energy
P	Maximum reported peak load
μ	Mean of Laplacian noise
$noise_t$	Calculated noise at that time interval

3.6. Masking with DPLM algorithm (calculation and addition of noise)

The phenomenon of preserving the privacy of real-time data is termed as masking in this paper. In order to initiate real-time data masking, a noise addition mechanism is required. After considering various noise addition mechanisms such as gaussian, exponential, and Laplacian mechanism, we consider Laplacian noise addition mechanism because it suits best when the output results are numerical [16]. Furthermore, in order to mask the original output, we added Laplacian noise in accumulated reading of energies at a single point of time t; as demonstrated in Definition 1. An illustrative definition of our proposed strategy is presented in Algorithm 1, and Fig. 3. According to Algorithm 1, we first accumulate the values of all energies ($Sum = SE + WE + GE$), this is done in order to calculate combined noise for all accumulated values. Afterwards, a function named "DiffNoiseCalculation" is called in the algorithm to initiate the process of noise calculation. In this function, first we calculate the value of maximum error that can be added or removed from the original value in order to perturb the reading. This error value is calculated using the term maximum error rate (%) which is determined on the basis of an agreement between the utility and smart meter using, which is used to calculate the amount of noise that is required to be added dynamically. After that, standard deviation (σ_{ε_0}) and variance ($\sigma^2_{\varepsilon_0}$) are calculated by using normal maximum probability (e.g. probability = 0.98) [7]. In all performed experiments, the particular value of differential privacy restriction variable ε in parameters σ_{ε_0} and $\sigma^2_{\varepsilon_0}$ is set to be 1 as in [17]. After calculation of standard deviation and variance, the scale b of Laplacian noise is calculated by using $\sigma^2_{\varepsilon_0}$ and total number of samples N. ($N = 1$ for one sample)

$$b = \sqrt{\frac{\sigma^2_{\varepsilon_0}}{2N}} \tag{5}$$

The values are then fed to Laplacian distribution which generate random noise by using probability density function given in Eq. (6).

$$Lap(x; \mu, b) = f(x; \mu, b) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}} \tag{6}$$

The above equation (6) can be simplified further by substituting the value of b from (5) to (6).

$$f\left(x;\mu,\sqrt{\frac{\sigma_{\varepsilon_0}^2}{2N}}\right)=\frac{1}{2\sqrt{\frac{\sigma_{\varepsilon_0}^2}{2N}}}.e^{\left(-\frac{|x-\mu|}{\sqrt{\frac{\sigma_{\varepsilon_0}^2}{2N}}}\right)} \tag{7}$$

For point-wise privacy, we take only one number of sample at that specific point ($N = 1$), so the above equation becomes:

$$f\left(x;\mu,0.707\sqrt{\sigma_{\varepsilon_0}^2}\right)=\frac{\sqrt{2}}{2\sqrt{\sigma_{\varepsilon_0}^2}}.e^{\left(-\frac{\sqrt{2}|x-\mu|}{\sqrt{\sigma_{\varepsilon_0}^2}}\right)} \tag{8}$$

By using mathematical identity to eradicate unnecessary roots, we get the following result:

$$f\left(x;\mu,0.707\sqrt{\sigma_{\varepsilon_0}^2}\right)=\frac{1}{(\sqrt{2})\sigma_{\varepsilon_0}}.e^{\left(-\frac{\sqrt{2}|x-\mu|}{\sqrt{\sigma_{\varepsilon_0}^2}}\right)} \tag{9}$$

If we take mean value of "$\mu = 0$" for Laplacian distribution, then Eq. (9) becomes:

$$f\left(x;0,\frac{1}{\sqrt{2}}\sqrt{\sigma_{\varepsilon_0}^2}\right)=\frac{1}{(\sqrt{2})\sigma_{\varepsilon_0}}.e^{\left(-\frac{\sqrt{2}|x|}{\sqrt{\sigma_{\varepsilon_0}^2}}\right)} \tag{10}$$

Finally, after substituting the value of Eq. (10) in Eq. (6), we get:

$$f(x;\mu,b)=\frac{1}{(\sqrt{2})\sigma_{\varepsilon_0}}.e^{\left(-\frac{\sqrt{2}|x|}{\sqrt{\sigma_{\varepsilon_0}^2}}\right)} \tag{11}$$

Laplacian distribution of Eq. (11) is further used to calculate appropriate noise which is sent further to the next step. In the next step, the accumulated value is masked by adding the calculated Laplacian noise by using the equation below:

$$SN_t = Sum_t + noise_t \tag{12}$$

According to step 15 of Algorithm 1, in order to normalize high peak values in the load, if any accumulated noise energy value exceeds the peak limit P of our approach (lets say 1800Wh), then to preserve additional privacy, P is subtracted from SN and is stored in a variable EX which is accumulated in the next reading and again this accumulated value is checked before transmission.

$$EX = SN - P \tag{13}$$

Finally, Algorithm 1 returns the final value for transmission to grid utility. Steps demonstrated in line 24–27 are used to calculate percentage error with respect to original value. This percentage error is accumulated and is checked at the end of each month to calculate demographics. We also added a table (Table 3) of percentage errors for different peak values, which is further discussed in Section 4. This inclusion of peak value will not have much effect over future load forecasting or demand response calculations, as the peak factor will be decided via agreement between smart meter user and grid utility. Therefore, utility will make necessary changes in their system in order to synchronize load forecasting and demand response with peak normalization.

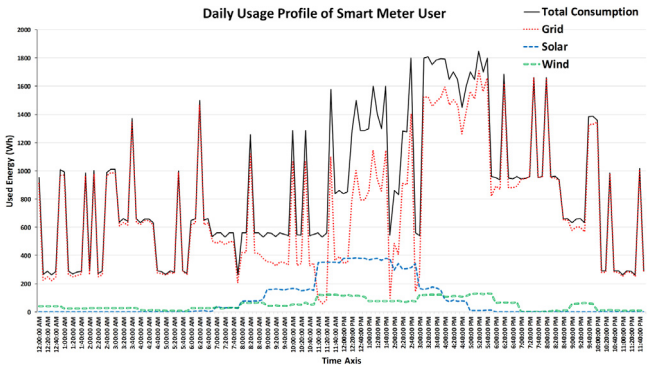


Fig. 4. Real time Data of Grid, Solar, and Wind energies for evaluation of Differential Privacy based Load Monitoring (DPLM) approach.

3.7. Privacy-utility trade-off from smart metering perspective

Differential privacy protects user data by perturbing it with appropriate amount of noise. As in our DPLM strategy, this noise is calculated using Laplacian distribution. However, this noise also has certain drawbacks, e.g. one of the major drawback is the reduction in data utility. In smart metering domain, the data from smart meters is used by grid utility for various purposes such as calculating demand response, perform load forecasting, etc. However, if the data perturb to a very large extent in order to protect privacy, then it can become useless for grid utility. Therefore, a precise and accurate borderline for trade-off is required before noise addition in order to preserve the privacy along with providing enough data utility [31]. In our DPLM strategy, we work over the notion of percentage error, which serves as the controlling factor of this trade-off. The value of percentage error is further fed to various parts of DPLM strategy which is used to calculate optimal noise data. His optimal percentage error is determined via mutual agreement between grid utility and smart meter users, and afterwards grid utility align and set their demand response and load forecasting calculations according to the error rate.

Similarly, there is no optimal trade-off value that works for every domain, as the requirement do vary according to scenario. For example, in some areas, the smart home users might be interested in preserving privacy to a large extent, in such areas the error rate can be reduced to minimum. However, in other areas, the smart meter users could be very curious regarding their privacy level, then their error rate can be increased to maximum possible limit and this maximum possible limit is determined via grid utility according to their requirement.

4. Performance evaluation of DPLM strategy

In this section, the performance evaluation and comparison of our proposed DPLM privacy preserving strategy has been carried out in a scenario of smart homes integrating RERs. The parameters we consider for evaluation of DPLM include real time data perturbation and percentage error. For implementation of smart metering scenario, we used grid energy consumption data from [37] and modified it according to RERs availability, and lastly we used the values of solar and wind energy similar as of [44]. We compare our results with the gaussian noise differential privacy (GNDP) approach proposed in [41]. For experimental evaluation, we used NumPy v1.14 library in Python 3.0 and iterated the data for 31 days with an interval of 10 min in each transmitted reading. The graph of experimental data is shown in Fig. 4. Furthermore, output results are generated for different peak values

Table 3
Monthly output results containing the values of total used energy, masked energy and percentage error.

Peak value (Wh)	Total energy (Wh)	Masked value (Wh)	Percentage error (%)
800	4844038	5075727	4.783
1000	4844038	4656531	3.87
1200	4844038	4916716	1.5
1500	4844038	4821676	0.461
2000	4844038	4834687	0.193

Table 4
Monthly accumulated readings for grid, solar, and wind energies.

Parameter	Description	Value (KWh)
TE	Total Energy	4844
TGE	Total Grid Energy	4218
TSE	Total Solar Energy	392
TWE	Total Wind Energy	234

and are further compared and classified according to evaluation parameters.

The transmission of real time data to grid utility is one of the major role of a smart meter. On the other hand, privacy of smart meter user also needs to be preserved. In order to judge the performance of DPLM privacy preserving strategy, we took data with 10 min interval for solar, wind, and grid utilization as presented in Fig. 4. Firstly, basic accumulation is applied to real-time values of solar, wind, and grid energies and are then stored in different variables as shown in Algorithm 1. After that, maximum added error rate (ME) (10%), standard deviation, variance, scale and point-wise noise is calculated. For example, lets take a random instantaneous reading of 500 Wh from each energy resource (solar, wind, and grid). The accumulated value will become 1500 Wh. The maximum error rate of 10% that is allowed by the utility is calculated as 150 Wh. The standard deviation (σ_{ε_0}) and the variance ($\sigma_{\varepsilon_0}^2$) of normal distribution for high probability value (e.g. 0.98) will be ($\sigma_{\varepsilon_0} = 63.57$) and ($\sigma_{\varepsilon_0}^2 = 4144$) respectively. Similarly, the scale can be calculated as:

$$b = \sqrt{\sigma_{\varepsilon_0}^2/(2N)} = \sqrt{4122/2(1)} = 45.52 \tag{14}$$

This value is further used to calculate the value of random Laplacian noise by using Eq. (11). Similarly, we used the parameter of peak value to restrict the data to a specific energy limit. For example, if the peak value is set to be 1000 Wh, then the maximum instantaneous value that can be transmitted after addition of Laplacian noise would be 1000 Wh. In this paper, we used peak values of 800, 1000, 1200, 1500, and 2000 Wh. According to output results, there comes a trade-off between the value of privacy and the value of accuracy that can be adjusted by changing values of maximum allowed error rate (ME) and peak point (P). In Fig. 5, we can see output graphs for 800, 1000, 1200, 1500, and 2000 peaks respectively. The peak values are determined on the basis of average electricity usage trend of smart meter user, and these values can vary according to the requirement and usage. Masking of smart meter data refers as preserving the original data and transmitting the new perturbed and protected value. Similarly, in presented graphs, masked values are the ones that we protected after adding noise and after restricting them to a maximum peak, and unmasked values are the original measured values of smart meter. As by visualizing the trend presented in the figure, we can say that with the increase in peak value, the more accurate data is being transmitted to utility. Similarly, deviation from different lower and upper peaks of original data because of addition of noise can also be observed in the presented output figures. For example, if point-wise value of energy is 500 Wh and the maximum error rate is 10%, then the output after perturbation can range from 450–550 Wh. And the utility or any intruder cannot guess that the transmitted value is the higher bound of any less value or the lower bound of any larger value. By keeping in view the all output graphs, we can conclude that the peak value of 1200 Wh with 10% error rate gives comparably desirable results by providing a moderate trade-off between privacy and accuracy according to our dataset. For comparison, we took idea of addition of gaussian noise from [41] as presented in Fig. 5(f). Finally, in Fig. 6, the comparison of unmasked value, values with gaussian noise, and values with DPLM strategy with 1200 Wh peak is presented. As it can be seen that our approach DPLM provides an optimal solution to transmit real time smart meter data by preserving the privacy of smart meter users.

4.1. Percentage error and monthly billing

To provide statistical data to utility, we formulate our algorithm to calculate the aggregated values of energies for a specific spell of time (let us say 31 days or a month). To check the accuracy of each point-wise transmitted reading after addition of noise, we calculate percentage error at end of each month. We compare the actual value of energy with the transmitted values and then calculated percentage error by using the ratio. The output values for different peak values are shown in Table 3. For example, in case of 800 Wh peak, the original value is 4844 *KWh*, while the output of transmitted accumulation is 5075 *KWh*. The calculation of these two values gives a percentage error of 4.783%, that is less than the value of maximum allowed error (10%). By analyzing the values in Table 3, it can be observed that the value of error reduces with increase in peak numbers, this happens because in case of higher peak value, mostly the accumulated data after noise addition is transmitted directly to utility and no value is stored in parameter (EX) for next calculation. Similarly, for future demand response and statistical calculation of solar, and wind data, we accumulate the values of each individual energy generated throughout the month. Moreover, for monthly billing, the value of grid usage for given time period is accumulated. These accumulated values are then transmitted to grid utility at the end of a specific time interval (e.g. 31 days) for calculation and billing purposes. The inclusion of noise will not have any effect over monthly billing, and user will only pay the amount of electricity they used from grid energy. Table 4 presents the name, description, and actual energy values for the output parameters.

4.2. Practicality of DPLM strategy from smart meter user perspective

From experimental evaluation we showed that our proposed DPLM strategy efficiently preserves smart meter user privacy. Similarly, we tried to develop a light-weight privacy preservation strategy so that it can easily be incorporated in every small meter, irrespective of its computational power. As it can be seen from experimental graphs, that the output of privacy preservation strategy strongly depends upon two important parameters: (1) maximum error rate (%), which determines the amount of noise that can be added. This parameter serves as a controlling factor and the grid utility adjust their calculation standards in accordance with this parameter value. (2) Peak factor (P), this parameter blocks reporting of peak load value, for example if the value of P is set to be 1000 Wh, and the accumulated value

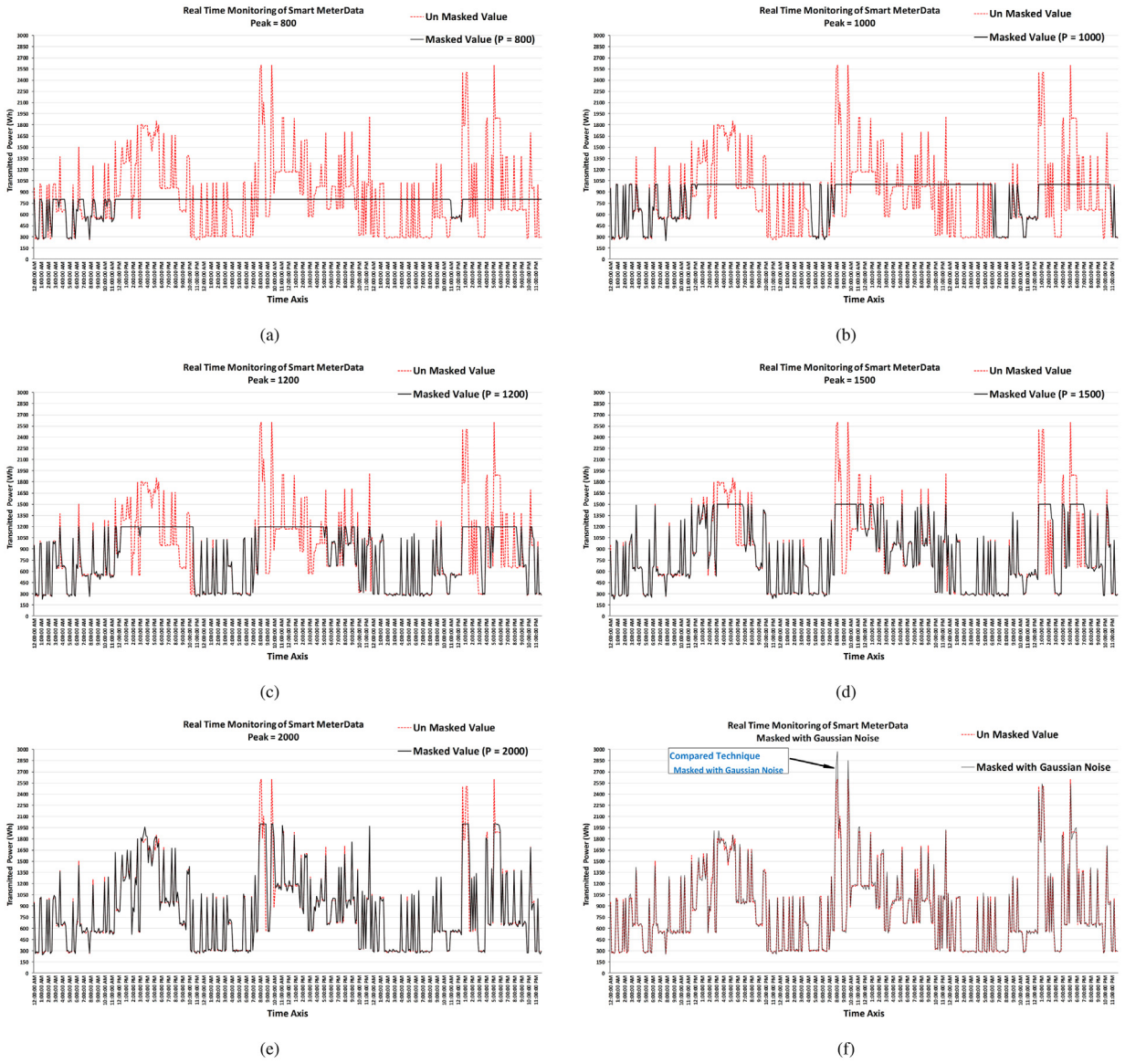


Fig. 5. Evaluation of masking approaches in renewable energy resources based smart grid scenario (a) DPLM Masking with P = 800 (b) DPLM Masking with P = 1000 (c) DPLM Masking with P = 1200 (d) DPLM Masking with P = 1500 (e) DPLM Masking with P = 2000 (f) Masking using Gaussian Noise [41].

after noise addition becomes 1500 Wh, then it filters out the extra 500 Wh and add only transmits 1000 Wh, then this surplus 500Wh is added in the next reading and the peak is checked again. Therefore, this peak protection normalizes the load graph and protects notable leakage of smart meter privacy. In order to practically implement DPLM strategy, a mutual agreement between grid utility and smart meter user is required so that users are satisfied with their privacy level and grid utility tune their demand response and load forecasting algorithms according to the output level in order to get maximum useful benefit from the data.

Furthermore, while developing DPLM strategy we consider three types of attacks that are commonly used to intrude into privacy of smart meter users. First and the foremost attack is NILM attack [28], in which the reported data of smart meter is used to predict the usage of specific appliance in smart home. In DPLM strategy, the specific peak normalizing factor blocks NILM attacks in such a way that these attacks could not predict the usage of any heavy appliance at a specific time, because the peak value is not guessable. Similarly, in case of low usage times, differentially private noise serves as a survivor of privacy that

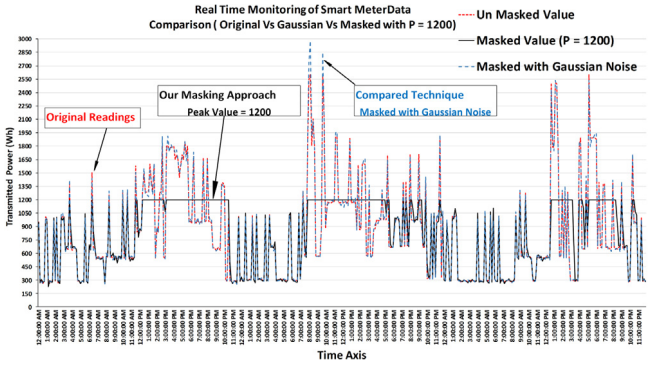


Fig. 6. Output Results Comparison (Unmasked Original Data **vs** DPLM Masking with P = 1200 **vs** Masking with Gaussian Noise Differential Privacy (GNDP) [41]).

makes it indistinguishable that whether which light appliance is being used. Second attack that we tackle is to overcome prediction of daily RER generation pattern. For example, from RER generation data, it can easily be predicted that at which time

solar panel was generating energy, and at which specific time wind was providing energy via turbine. This predicted data can further be used by grid utility to increase price of electricity at a time when RERs are not available for generation, similarly, this data may also increase social privacy issues. Therefore, DPLM ensures that daily and monthly output data does not reveal any generation pattern. Finally, the third attack that we considered was the weekly schedule monitoring attack over smart home residents. In this attack, the weekly accumulated data of smart meter is compared with data of previous weeks in order to predict the weekly behavior. However, our DPLM effectively protects privacy leakage in case of such attacks by efficiently using the phenomenon of maximum error and peak load protection.

Hence, by analyzing output results of Figs. 5 and 6, and by detailed discussion about practicality, we can say that our proposed DPLM strategy efficiently protects the privacy of smart meter users by limiting peak values and perturbing the data by using Laplacian noise mechanism effectively. In this way, the utility cannot estimate both; the availability of specific RER at any particular time and the lifestyle of smart meter users. However, the transmitted data can still be used for future demand response and load forecasting calculations.

5. Conclusion

With the increase in prices of traditional fossil fuels, the trend of integration of renewable energy resources (RERs) in smart homes is considered to be the future of smart grid technology. With the advancement in information and communication technologies, and sensing techniques, the traditional energy grid is becoming smart. The smart grid is considered to be smart when it includes various modern technologies, such as demand response calculation, load forecasting, and real time data monitoring. Similarly, these modernized technologies can be a severe hazard to personal privacy of smart meter users because of real-time monitored data. In this paper, a differential privacy load monitoring (DPLM) mechanism has been proposed to preserve private information of smart meter users. Our developed approach masks real time data in such a way that it does not reveal the personal lifestyle and availability of RERs, and on the other hand it provides adequate information to grid utility that can be used for calculation of demand response and load forecasts. After experimental evaluation and comparison with GNDP approach, we concluded that DPLM strategy provides an impressive and state of the art solution to overcome privacy issues in RERs based smart homes by efficient data perturbation, protecting peak values, and having only 1.5% error at peak value of 1200. As a part of future work, we will be working on security and privacy protection of different scenarios of smart grid by enhancing differential privacy and combining it with different security and privacy protocols.

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Conflict of interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.jpdc.2019.04.012.

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